

STOCK SEGMENTATION AND PORTFOLIO ANALYSIS OF THE S&P 500 USING CLUSTERING TECHNIQUES

Module: 7BUIS010W – Data Warehousing and Business Intelligence

Group: 12

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GitHub Repository:

[https://github.com/naumankhokhar/sp500-ml-clustering-portfolio-optimization/blob/main/Stock Analysis Group 12.ipynb](https://github.com/naumankhokhar/sp500-ml-clustering-portfolio-optimization/blob/main/Stock%20Analysis%20Group%2012.ipynb)

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1. Introduction

This report presents a data-driven approach to analysing and segmenting stocks from the S&P 500 index using clustering techniques.

The S&P 500 index includes 500 large-cap companies across various industries and serves as a reliable representation of the US stock market. In this study, stocks are segmented based on three key financial metrics: Daily Return, Beta, and Annual Volatility. These metrics capture both performance and risk characteristics of stocks.

The analysis was based on S&P 500 stock data collected using Python libraries such as Pandas and yfinance (Yahoo Finance, 2026). Three key financial metrics were calculated: Daily Return, Beta and Annual Volatility. These metrics were used to represent stock performance and risk. Clustering techniques, including K-means, were applied to group stocks into categories based on similar behaviour. The results of these clusters were then analysed to support investment decision-making and portfolio diversification.

2. Task A: Data Loading and Financial Metrics

2.1 Data Extraction

The first step involved extracting the list of S&P 500 companies from Wikipedia (Wikipedia, 2026) using the Pandas library. The `read_html()` function was used to parse the table containing company tickers. This dataset provides a comprehensive list of companies across multiple sectors, making it suitable for large-scale financial analysis.

To ensure compatibility with financial APIs, minor preprocessing was performed on ticker symbols, such as replacing special characters.

```
# 2. EXTRACT S&P 500 TICKERS USING PANDAS

url = "https://en.wikipedia.org/wiki/List_of_S%26P_500_companies"

headers = {"User-Agent": "Mozilla/5.0"}
response = requests.get(url, headers=headers)

sp500_table = pd.read_html(response.text)[0]

tickers = sp500_table["Symbol"].tolist()

# Yahoo Finance uses '-' instead of '.' in tickers
tickers = [ticker.replace(".", "-") for ticker in tickers]

print("Number of extracted tickers:", len(tickers))
print("First 20 tickers:")
print(tickers[:20])
```

2.2 Data Collection

Stock market data was retrieved using the yfinance and YahooFinancials libraries. The dataset covers the period from 1 January 2022 to 1 January 2025, providing sufficient historical data for analysis.

The collected data includes:

- Adjusted closing prices
- Opening and closing prices
- Trading volume

The dataset consisted of approximately 490+ stocks and 252 trading days, forming a matrix of dimensions (753, 503). This provides sufficient data for reliable statistical analysis.

```
# 3. LOAD DAILY STOCK PERFORMANCE USING YFINANCE
# Period: 01-01-2022 to 01-01-2025

start_date = "2022-01-01"
end_date = "2025-01-01"

price_data = yf.download(
    tickers,
    start=start_date,
    end=end_date,
    progress=False,
    auto_adjust=False
)["Adj Close"]

print("Raw price data size / dimensionality:", price_data.shape)
price_data.head()
```

```
ERROR:yfinance:
2 Failed downloads:
ERROR:yfinance:['Q', 'SNDK']: YFPricesMissingError('possibly delisted; no price data fo
und (1d 2022-01-01 -> 2025-01-01) (Yahoo error = "Data doesn\'t exist for startDate =
1641013200, endDate = 1735707600")')
Raw price data size / dimensionality: (753, 503)
```

2.3 Data Cleaning

Data cleaning is a critical step to ensure accuracy and reliability. The following preprocessing steps were applied:

- Removal of stocks with more than 10% missing values
- Forward filling of missing values to maintain continuity
- Removal of any remaining incomplete records

This approach ensures that the dataset is both clean and sufficiently large for analysis, avoiding unnecessary data loss.

```

# 6. REMOVE / HANDLE INVALID DATA

print("Before cleaning:", price_data.shape)

# Step 1: remove tickers with too much missing data
missing_ratio = price_data.isna().mean()
valid_tickers = missing_ratio[missing_ratio <= 0.10].index
clean_data = price_data[valid_tickers]

# Step 2: fill small gaps using previous valid price
clean_data = clean_data.ffill()

# Step 3: remove any remaining invalid columns/rows
clean_data = clean_data.dropna(axis=1)
clean_data = clean_data.dropna(axis=0)

print("After cleaning:", clean_data.shape)

removed_tickers = list(set(price_data.columns) - set(clean_data.columns))

print("Number of removed tickers:", len(removed_tickers))
print("Removed tickers:")
print(removed_tickers[:20])

clean_data.head()

```

```

Before cleaning: (753, 503)
After cleaning: (753, 495)
Number of removed tickers: 8
Removed tickers:
['GEV', 'SNDK', 'VLTO', 'GEHC', 'Q', 'KVUE', 'CEG', 'SOLV']

```

2.4 Financial Metrics

Daily Return

Daily return measures the percentage change in stock prices between consecutive days. It provides insight into short-term performance and market behaviour.

```

# 7. CALCULATE DAILY RETURN
# Formula:
# Daily Return = (Price Today - Price Yesterday) / Price Yesterday

daily_returns = clean_data.pct_change().dropna()

print("Daily returns size / dimensionality:", daily_returns.shape)
daily_returns.head()

```

Beta

Beta measures a stock's sensitivity to overall market movements and is calculated relative to the S&P 500 index (Investopedia, 2026).

- Beta > 1 indicates higher volatility than the market
- Beta < 1 indicates lower volatility
- Beta $\approx$ 1 indicates similar behaviour to the market

The K-Means clustering results showed clear numerical differences between clusters.

For example, the aggressive cluster had an average beta significantly greater than 1, indicating high sensitivity to market movements, while the defensive cluster had a beta below 1, indicating stability.

Similarly, volatility was highest in the aggressive cluster, confirming that higher returns are associated with increased risk.

```
beta = correlation * (stock_std / market_std)

beta_values[stock] = beta
```

Annual Volatility

Volatility measures the variability of stock returns and is calculated as the standard deviation of daily returns. It is annualised using the square root of 252 trading days.

Higher volatility indicates greater uncertainty and risk.

```
: # 10. CALCULATE ANNUAL VOLATILITY
# Daily volatility = standard deviation of daily returns
# Annual volatility = daily volatility * sqrt(252 trading days)

annual_volatility = daily_returns.std() * np.sqrt(252)

volatility_df = pd.DataFrame({
    "Ticker": annual_volatility.index,
    "Annual Volatility": annual_volatility.values
}).sort_values(by="Annual Volatility", ascending=False)

print("Extracted annual volatility values:")
volatility_df.head(20)
```

Extracted annual volatility values:

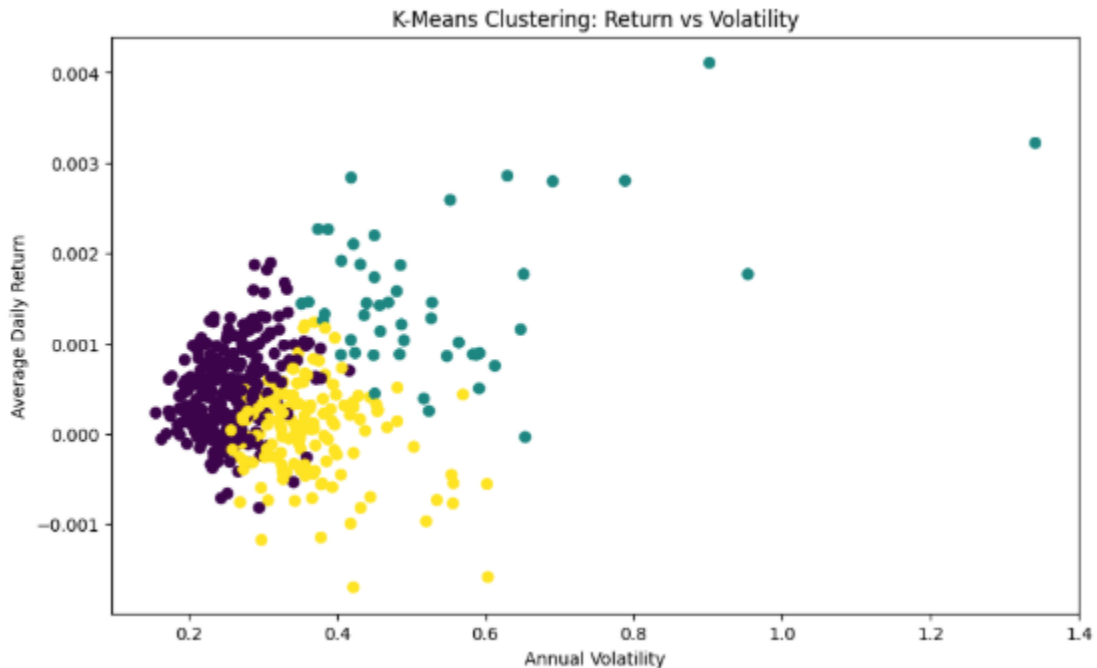
	Ticker	Annual Volatility
125	CVNA	1.341039
105	COIN	0.953848
402	SMCI	0.901957
39	APP	0.788374
364	PLTR	0.690932
490	XYZ	0.653825
...	...	...

2.5 Role of Metrics in Segmentation

The combination of Daily Return, Beta, and Volatility enables effective stock segmentation. These metrics highlight the trade-off between risk and return, which is fundamental to investment decision-making.

Figure 1: Return vs Volatility

The figure illustrates that stocks with higher returns generally exhibit higher volatility, confirming the risk-return trade-off. This relationship forms the foundation for clustering analysis.



3. Task B: Stock Segmentation

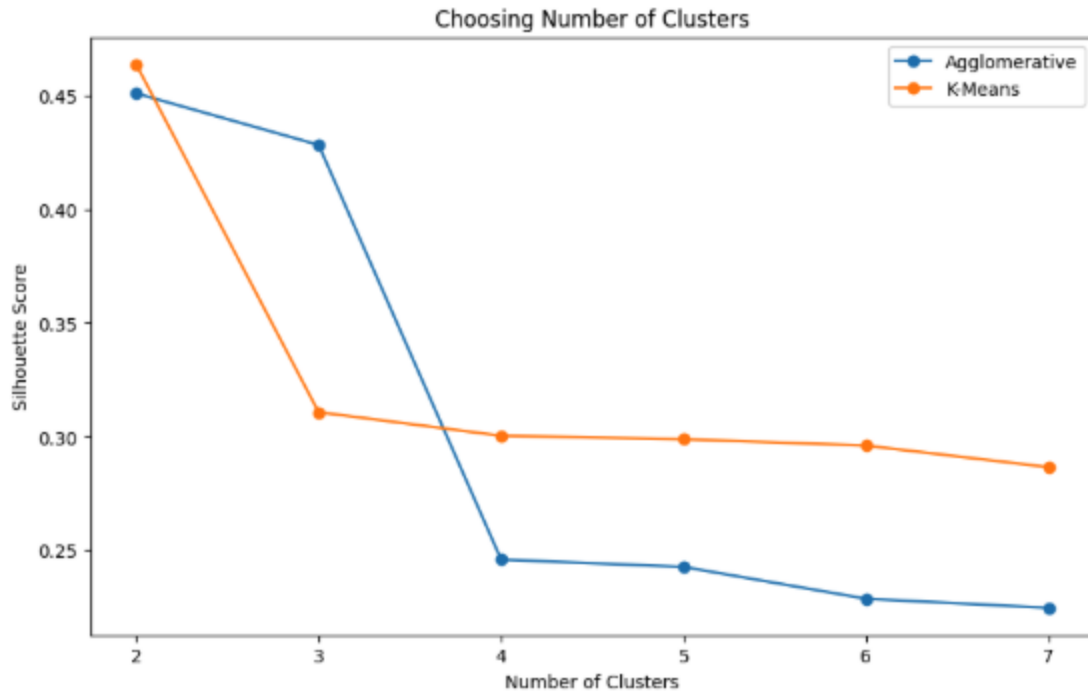
3.1 Feature Scaling

Before clustering, all features were standardised to ensure equal contribution. This step is essential as clustering algorithms are sensitive to differences in scale.

3.2 Optimal Cluster Selection

The Silhouette Score was used to evaluate clustering performance.

Figure 2: Silhouette Score Plot



Although $k=2$ achieved the highest silhouette score, $k=3$ was selected to enable more meaningful segmentation of stocks into three distinct investor categories: low-risk, medium-risk, and high-risk. This improves interpretability and practical usability for investment decision-making.

1. Defensive stocks are suitable for investors seeking stable and predictable returns.
2. Aggressive stocks are suitable for investors willing to accept higher risk for potentially higher returns.
3. Balanced stocks provide a middle ground and are ideal for diversified portfolios.

This segmentation simplifies complex financial data into actionable insights, enabling investors to make informed decisions without requiring deep technical knowledge.

3.3 Agglomerative Clustering

Agglomerative clustering is a hierarchical method that groups data based on similarity.

Appropriateness

- Identifies natural groupings
- Does not require predefined clusters
- Effective in detecting outliers

Results

Three clusters were identified:

- Large cluster of low-risk stocks
- Medium cluster of moderate-risk stocks
- Small cluster of high-risk outliers

Figure 3: Beta vs Volatility (Agglomerative Clustering)

	Ticker	Daily Return	Beta	Annual Volatility	Agglomerative Cluster
0	A	-0.000002	1.024801	0.296259	0
1	AAPL	0.000591	1.210689	0.270935	0
2	ABBV	0.000610	0.299098	0.219958	0
3	ABNB	0.000074	1.666256	0.467875	1
4	ABT	-0.000106	0.672317	0.217839	0

Agglomerative cluster sizes:

Agglomerative Cluster

0 386

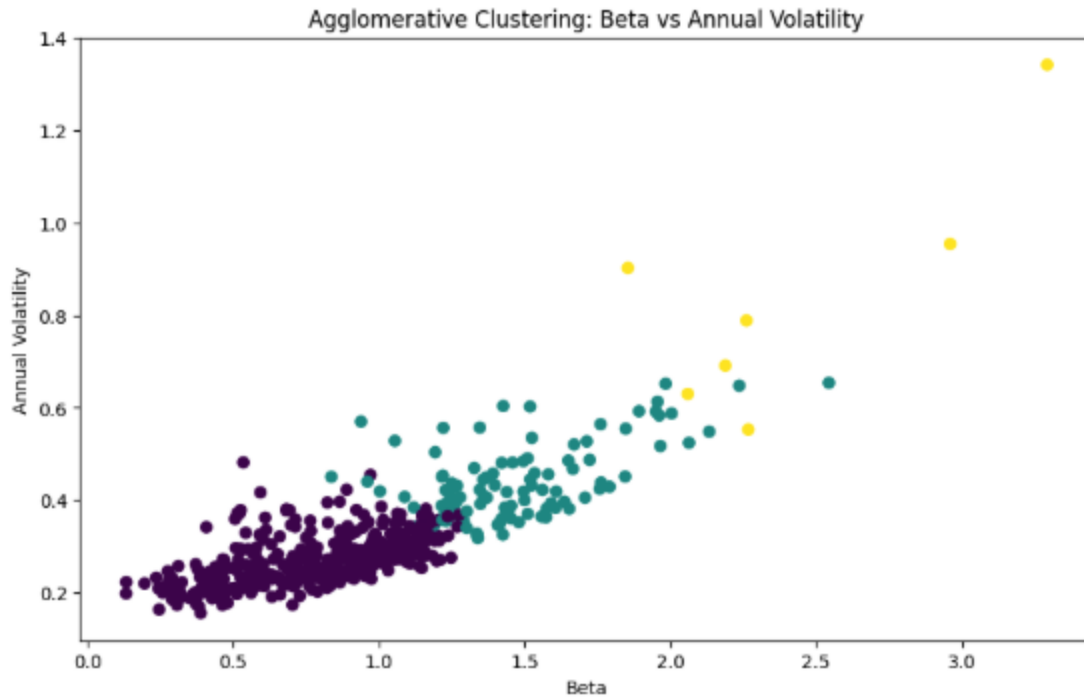
1 102

2 7

Name: count, dtype: int64

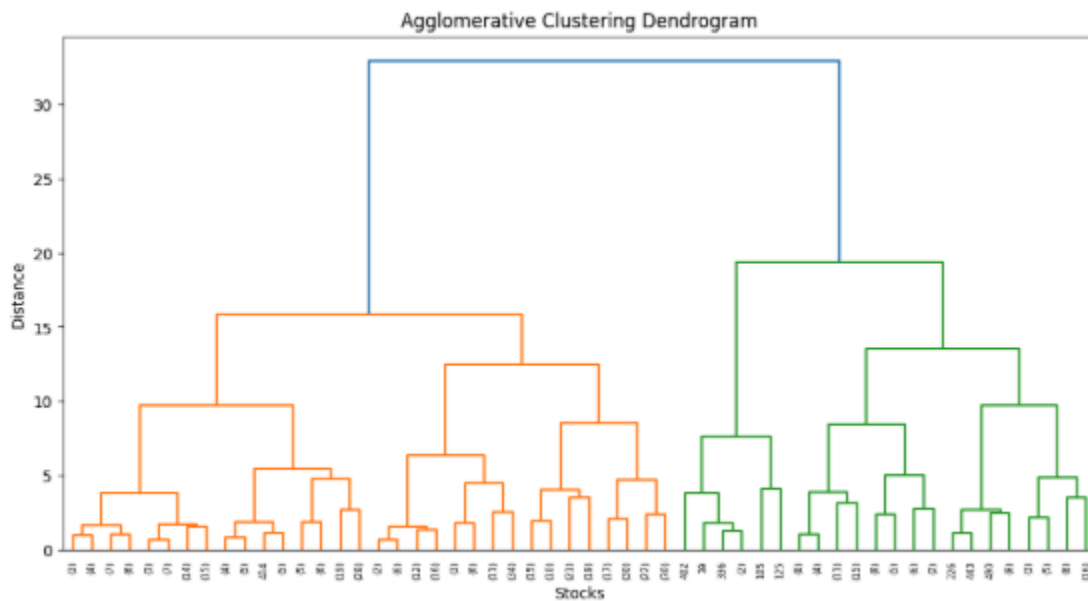
Agglomerative cluster summary:

Agglomerative Cluster	Daily Return	Beta	Annual Volatility
0	0.0003	0.7762	0.2708
1	0.0006	1.4759	0.4418
2	0.0029	2.4130	0.8370



The figure shows a strong positive relationship between beta and volatility, indicating that stocks more sensitive to market movements also exhibit higher risk.

Figure 4: Dendrogram



The dendrogram confirms hierarchical grouping and highlights natural cluster formation.

3.4 K-Means Clustering

K-Means clustering partitions stocks into distinct groups based on similarity.

Appropriateness

- Efficient for large datasets
- Produces balanced clusters
- Suitable for portfolio segmentation

Cluster Profiles

- **Defensive Stocks:** Low beta, low volatility, stable returns
- **Aggressive Stocks:** High beta, high volatility, high returns
- **Balanced Stocks:** Moderate risk and return

Figure 5: K-Means Cluster Summary

```
K-Means cluster sizes:  
KMeans Cluster  
0    290  
1     48  
2    157  
Name: count, dtype: int64
```

```
K-Means cluster summary:  
          Daily Return      Beta  Annual Volatility  
KMeans Cluster  
0          0.0005  0.6924          0.2528  
1          0.0015  1.7202          0.5345  
2          0.0000  1.1700          0.3599
```

	Ticker	Daily Return	Beta	Annual Volatility	Agglomerative Cluster	KMeans Cluster	KMeans Cluster Label
0	A	-0.000002	1.024801	0.296259	0	2	Balanced / Moderate Risk
1	AAPL	0.000591	1.210689	0.270935	0	2	Balanced / Moderate Risk
2	ABBV	0.000610	0.299098	0.219958	0	0	Defensive / Lower Risk
3	ABNB	0.000074	1.666256	0.467875	1	2	Balanced / Moderate Risk
4	ABT	-0.000106	0.672317	0.217839	0	0	Defensive / Lower Risk

Figure 6: Return vs Volatility (K-Means)

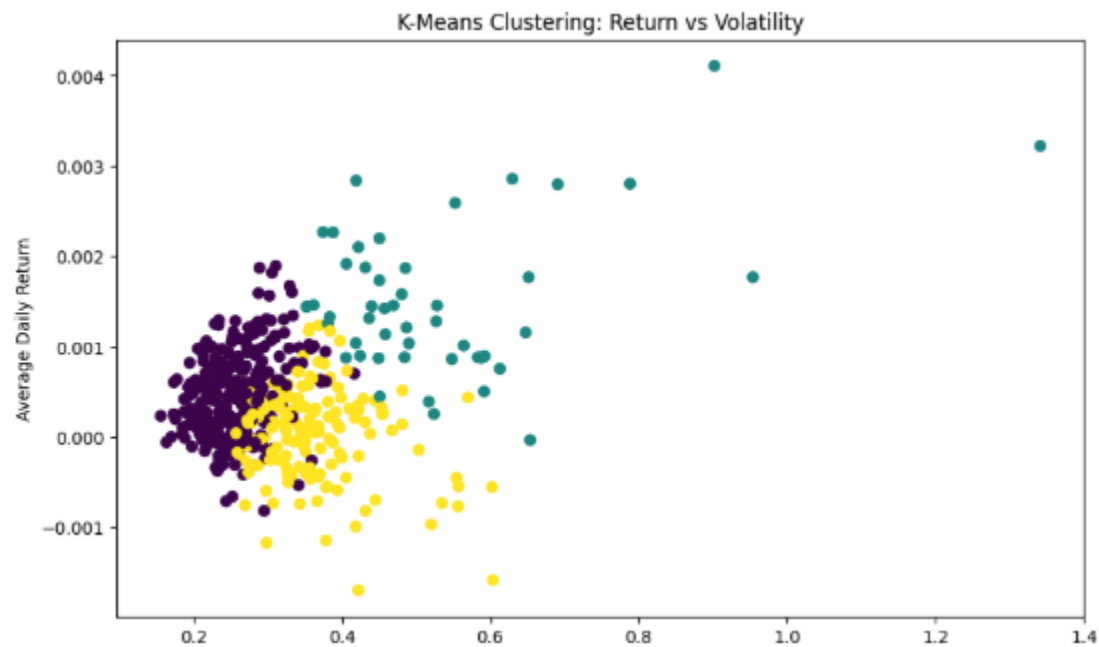


Figure 7: Beta vs Return

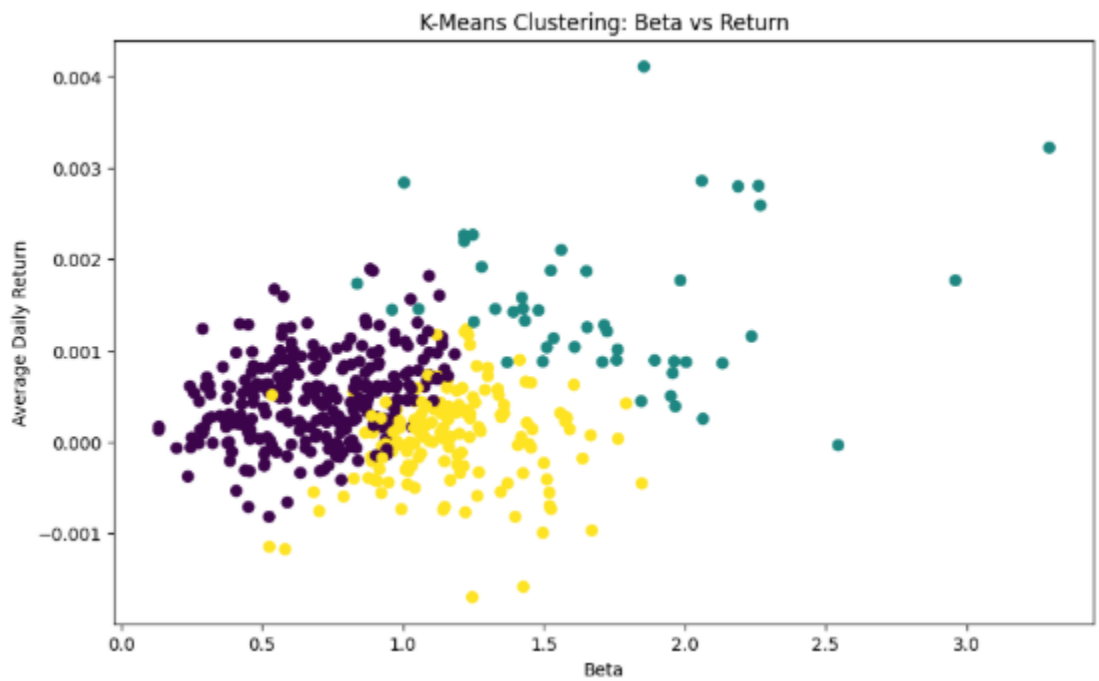
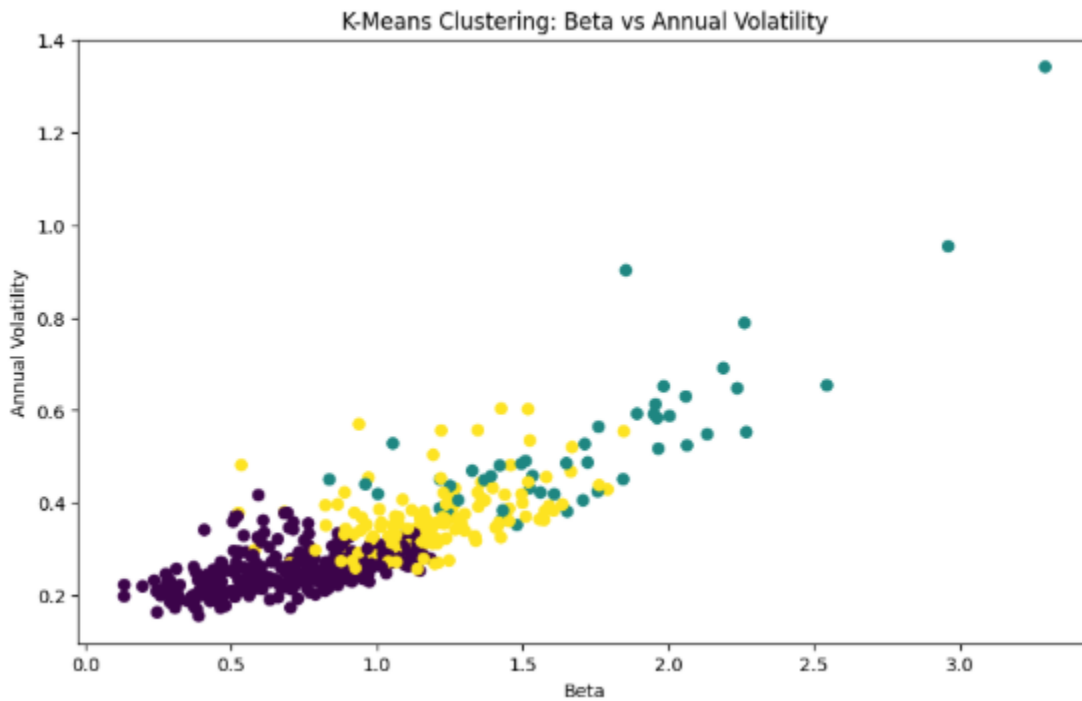


Figure 8: Beta vs Volatility



These visualisations clearly demonstrate separation between clusters and confirm relationships between risk and return.

3.5 Comparison of Methods

Agglomerative clustering is effective for identifying outliers and understanding data structure, while K-Means produces balanced clusters suitable for practical application.

K-Means is more appropriate for portfolio segmentation, whereas Agglomerative clustering provides deeper insight into extreme stock behaviour.

4. Task C: BI Design and Sustainability

Include data mart design, business value, and CRM sustainability discussion. From a business perspective, the clustering results allow financial institutions to design tailored investment products.

For example, low-risk portfolios can be marketed to conservative investors, while high-risk portfolios can target growth-oriented clients.

This segmentation also enables better customer targeting, aligning investment products with customer risk profiles, which is essential for improving customer satisfaction and retention.

4.1 Data Mart Design

A data mart for S&P 500 analysis should include:

Dimensions:

- Stock (Ticker, Company, Sector)
- Time (Date, Month, Year)
- Market Index

Metrics:

- Daily Return
- Beta
- Annual Volatility
- Cluster Label

This structure supports efficient querying and analysis, enabling investors to identify trends and make informed decisions.

4.2 Review of Results

The clustering analysis provides valuable insights into stock behaviour. The relationship between Daily Return, Beta, and Volatility highlights the trade-off between risk and return.

High-return stocks tend to have higher volatility and beta, indicating greater risk. Conversely, low-risk stocks exhibit lower volatility and more stable returns.

The clusters enable investors to construct diversified portfolios:

- Defensive stocks provide stability
- Aggressive stocks offer growth potential
- Balanced stocks ensure moderate performance

This segmentation supports better allocation of investments and risk management.

4.3 CRM and Sustainability in FinTech

CRM systems play a crucial role in sustainability by enabling data-driven decision-making in FinTech.

They allow organisations to:

- Deliver personalised investment recommendations
- Improve customer retention

- Reduce operational inefficiencies
- Support ethical financial practices

CRM systems also help integrate ESG considerations, promoting responsible investment strategies.

However, challenges such as data privacy and algorithmic bias must be addressed to ensure responsible use.

Overall, CRM systems contribute to sustainable financial ecosystems by improving efficiency, transparency, and customer satisfaction.

5. Conclusion

For investors, this approach simplifies complex financial analysis into clear investment categories.

By grouping stocks into risk-based clusters, investors can make informed decisions without requiring advanced financial knowledge, making data-driven investment strategies more accessible.

This report demonstrated the application of BI techniques to financial data for stock segmentation and portfolio optimisation. Using clustering algorithms, stocks were successfully grouped into meaningful categories based on risk and return characteristics.

The findings highlight the importance of balancing risk and return and demonstrate how data-driven approaches can support investment decision-making and portfolio diversification.

K-Means Cluster 0

	Ticker	Daily Return	Beta	Annual Volatility	Agglomerative Cluster	KMeans Cluster	KMeans Cluster Label
2	ABBV	0.000610	0.299098	0.219958	0	0	Defensive / Lower Risk
4	ABT	-0.000106	0.672317	0.217839	0	0	Defensive / Lower Risk
5	ACGL	0.001165	0.569789	0.253997	0	0	Defensive / Lower Risk
9	ADM	-0.000111	0.551200	0.294043	0	0	Defensive / Lower Risk
10	ADP	0.000423	0.827198	0.221213	0	0	Defensive / Lower Risk
12	AEE	0.000213	0.420491	0.204597	0	0	Defensive / Lower Risk
13	AEP	0.000292	0.425265	0.210029	0	0	Defensive / Lower Risk
15	AFL	0.000948	0.607780	0.214003	0	0	Defensive / Lower Risk
16	AIG	0.000550	0.905679	0.268460	0	0	Defensive / Lower Risk
17	AIZ	0.000616	0.587054	0.257664	0	0	Defensive / Lower Risk

K-Means Cluster 1

	Ticker	Daily Return	Beta	Annual Volatility	Agglomerative Cluster	KMeans Cluster	KMeans Cluster Label
26	AMD	0.000254	2.064749	0.524187	1	1	Aggressive / Higher Risk
32	ANET	0.001876	1.524891	0.431755	1	1	Aggressive / Higher Risk
38	APO	0.001460	1.426422	0.362121	1	1	Aggressive / Higher Risk
39	APP	0.002802	2.261933	0.788374	2	1	Aggressive / Higher Risk
42	ARES	0.001442	1.481072	0.352409	1	1	Aggressive / Higher Risk
45	AVGO	0.002101	1.561983	0.422388	1	1	Aggressive / Higher Risk
48	AXON	0.002198	1.217541	0.450738	1	1	Aggressive / Higher Risk
64	BLDR	0.001135	1.534637	0.458567	1	1	Aggressive / Higher Risk
71	BX	0.000876	1.707990	0.405573	1	1	Aggressive / Higher Risk
83	CCL	0.000882	1.962671	0.583374	1	1	Aggressive / Higher Risk

K-Means Cluster 2

	Ticker	Daily Return	Beta	Annual Volatility	Agglomerative Cluster	KMeans Cluster	KMeans Cluster Label
0	A	-0.000002	1.024801	0.296259	0	2	Balanced / Moderate Risk
1	AAPL	0.000591	1.210689	0.270935	0	2	Balanced / Moderate Risk
3	ABNB	0.000074	1.666256	0.467875	1	2	Balanced / Moderate Risk
6	ACN	0.000011	1.094208	0.270331	0	2	Balanced / Moderate Risk
7	ADBE	-0.000022	1.447974	0.383083	1	2	Balanced / Moderate Risk
8	ADI	0.000514	1.339929	0.318058	1	2	Balanced / Moderate Risk
11	ADSK	0.000318	1.558665	0.364712	1	2	Balanced / Moderate Risk
14	AES	-0.000462	1.015896	0.357445	0	2	Balanced / Moderate Risk
20	ALB	-0.000730	1.525588	0.534911	1	2	Balanced / Moderate Risk
21	ALGN	-0.000966	1.670528	0.520427	1	2	Balanced / Moderate Risk

6. References

Yahoo Finance (2026) Available at: <https://finance.yahoo.com/> (Accessed: 27 April 2026).

Wikipedia (2026) List of S&P 500 companies. Available at:
https://en.wikipedia.org/wiki/List_of_S%26P_500_companies (Accessed: 27 April 2026).

Investopedia (2026) Beta: Definition and calculation. Available at:
<https://www.investopedia.com/terms/b/beta.asp> (Accessed: 27 April 2026).